C 42961 **(Pages : 3)** **Name.....**

Reg. No.....

SECOND SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION, APRIL 2023

Physics/Applied Physics

PHY 2B 02/APY 2B 02—PROPERTIES OF MATTER WAVES AND ACOUSTICS

(2017–2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions.*
1 *mark each.*

1. Relation between Y, η and σ is ————————.
2. A horizontal beam fixed at one end and loaded at the other end is called ————————.
3. The number of vibrations in one second is called ————————.
4. What is a damped harmonic oscillator ?
5. What is meant by resonant frequency of a forced harmonic oscillator ?
6. In transverse wave motion, Particles of the ———————— vibrate ———————— the direction of wave propagation.
7. What is the unit of energy density ?
8. What are antinodes ?
9. Write the relation between velocity of sound and absolute temperature
10. What is reverberation ?

(10 × 1 = 10 marks)

Section B

Answer **all** *questions.*
Write in two or three sentenses.
2 *marks each*

11. Define Bulk modulus of elasticity with equation.
12. What is angle of twist ?
13. What do you mean by intensity of a wave ?
14. How 'I' section girders work ?

Turn over

15. Define Fourier's theorem.
16. Define Q factor.
17. Differentiate between wave velocity and group velocity.

(7 × 2 = 14 marks)

Section C

Write any **five** *questions.*
Write in one paragraph.
4 *marks each.*

18. Obtain the expression for velocity of sound wave.
19. Deduce the expression for couple per unit twist of a uniform solid cylinder.
20. Derive the expression for the depression at the loaded end of a cantilever.
21. Explain the terms : a) amplitude b) period c) frequency of wave motion.
22. Write down difference between reflections at a free boundary and rigid boundary.
23. Explain the modes of vibrations in stretched string.
24. Explain nodes and antinodes.

(5 × 4 = 20 marks)

Section D

Solve any **four** *problems.*
4 *marks each.*

25. Calculate the speed of transverse wave in a wire of radius 0.56 mm subjected to a tension of 2kgwt. Given the density of the material of the wire is 8500 kg/m^2.
26. A particle makes S.H.M. along a straight line and its velocity when passing through points 3 cm and 4 cm from the centre of its path is 16 cm/s and 12 cm/s respectively. Find (a) the amplitude, (b) the time period of motion.
27. Aparticle of mass 10kg lies in a potential field V = (50x^2 + 100) Joule/kg. Deduce the frequency of oscillation.
28. A light metal rod of length 60 cm and of radius 1 cm is clamped at one end loaded at the free end, with 5.5 kg. Calculate the depression of the free end assuming Y = 9 × 10^{10}N/m^2.
29. A circular disc of radius 20cm oscillates as a pendulum about a point on its circumference. Calculate the period of oscillation.

30. Find a Fourier's series for the function defined by the equations

    Y = 1 for t = 0 to t = T/2

    0 for t = T/3

    +1 for t = T/2 to t = T

31. Prove that the velocity of a particle in S.H.M. at a distance $\sqrt{3}/2$ of its amplitude from the centre is half of its velocity at the central position.

(4 × 4 =16 marks)

Section E

Write any **two**.
10 *marks each.*

32. Explain transverse waves in stretched string and derive the equation for velocity

33. Define Poisson's ratio. Derive the relation connecting Y, η and σ.

34. What is damping force ? Derive the equation for damped harmonic oscillator. Explain over damped and under damped cases.

35. Study the variation of KE, PE and Total energy of SHM with displacement.

(2 × 10 = 20 marks)